

Home Credit Scorecard Classification Model

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Final Task of Project-Based Internship Home Credit Indonesia x Rakamin Academy
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Problem Research

application_train.csv from PBI
HCI x Rakamin Academy

Context:

Home Credit Indonesia wants to build a model to predict credit score of their customers. The data contains 122 columns including 1 'TARGET' column.

Objectives

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Accelerate the credit score calculation by using machine learning model and avoid to reject potential customer.

Goals

- Understanding the characteristic of customer that has good/bad score
- Building machine learning model to predict credit score classification
- Providing business recommendation

Business Metrics

Metric: credit score ratio

Data Pre-Processing

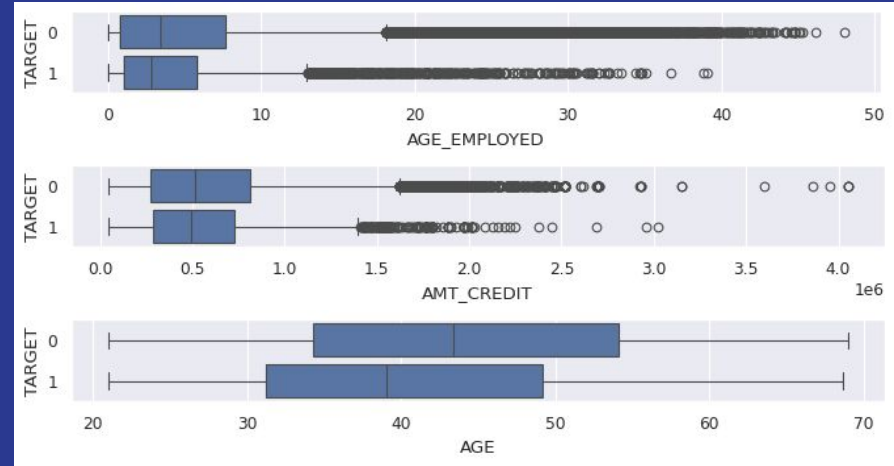
- Check the missing values, delete feature that has a lot of missing values.
- Do univariate analysis and clean the features.
- Change the 'days' column to age/year.
- Transform the binary categorical features to binary numerical.
- Check the correlation of numerical features.



- Feature Transformation: Use StandardScaler to non binary numerical features.
- Do encoding to selected categorical feature.
- Do oversampling to handle imbalanced data.

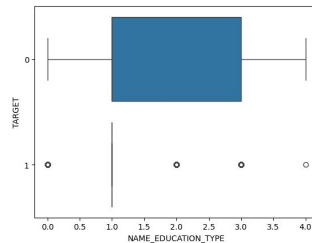
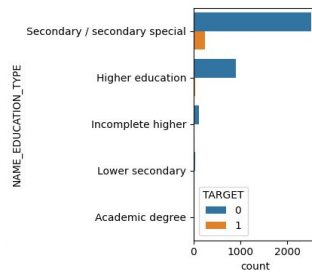
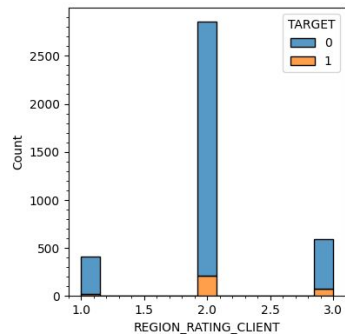
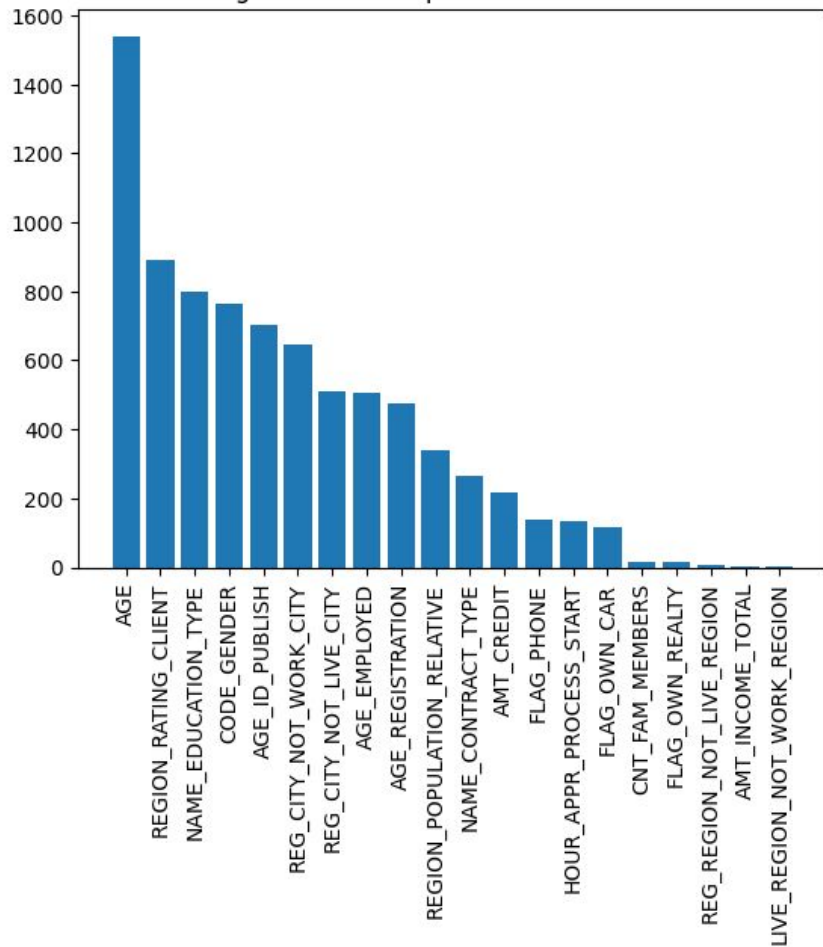
Data Visualization & Business Insight

Exploratory Data Analysis



- Target value: 0 good 1 bad credit scores.
- Customers with more amount of credit potentially have good scores.
- The same insight as age and age employed. Older people tend to have better credit scores.

Ranking of Feature Importance Based on ANOVA



Insights:

- Age is the most important feature by ANOVA method.
- Older customers are more likely to have better credit scores. It is possible because most older people are more mature, responsible, and financially stable.
- Another important features are region rating and education type.
- Customers with education higher than secondary/secondary special rarely have bad credit scores.

Machine Learning Implementation & Evaluation

Metode:

- Logistic Regression
- Decision Tree
- Hyperparameter Tuning

The results

Model	Before Hyperparameter Tuning		After Hyperparameter Tuning	
	Accuracy (train)	Accuracy (test)	Accuracy (train)	Accuracy (test)
Logistic Regression	65.735%	60.047%	65.735%	60.047%
Decision Tree	100%	82.464%	91.181%	87.877%

For this case, decision tree with hyperparameter tuning is the best prediction model.

Business Recommendation

Link:

<https://github.com/uniainiyah/HCI.git>

- For this data, a machine learning model using a Decision Tree gives better results.
- Older people, experienced professionals, and people with higher education are more likely to have good scores. Home credit can target this segment of customers to offer the product.